WINDOWS AUTHENTICATION LOG ANALYSIS

Detecting and Investigating Authentication Events Using Windows Security Logs

SOC Analyst Home Lab Project

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Project Overview

This report documents the investigation of Windows authentication events within a controlled cybersecurity home lab environment. The analysis focuses on identifying failed and successful login attempts using Windows Security Event Logs and evaluating the activity for indicators of suspicious behavior.

Skills Demonstrated

- Security Monitoring
- Authentication Analysis
- Event Log Investigation
- Threat Detection
- Incident Response Fundamentals
- Security Documentation

Classification: Educational Cybersecurity Project

Project Overview

This project demonstrates the analysis of Windows Security Event Logs to identify authentication-related activities, including failed login attempts and successful user logins. The objective was to investigate login events, determine whether suspicious activity occurred, and document the findings using a structured incident response approach.

Objective

The purpose of this investigation was to:

- Monitor Windows authentication events.
- Identify failed and successful login attempts.
- Analyze relevant Security Event Logs.
- Determine whether the observed activity was indicative of malicious behavior or normal user activity.
- Practice fundamental Security Operations Center (SOC) analyst skills including log analysis, event correlation, and incident documentation.

Lab Setup

Environment

Component	**Details**
Operating System	Windows 10 Virtual Machine
Analysis Tool	Windows Event Viewer
Log Source	Windows Security Logs
Attacker Simulation	Manual failed login attempts
Network Type	Virtual Lab Environment

Systems

Windows Machine

- IP Address: 192.168.20.10
- User Account: vboxuser

Kali Linux Machine

- IP Address: 192.168.20.11

The environment was configured within a home cybersecurity lab for educational and defensive security purposes.

Events Analyzed

The following Windows Security Event IDs were investigated:

Event ID 4625 – Failed Logon

This event is generated when a user authentication attempt fails.

Key fields reviewed:

- Account Name
- Logon Type
- Failure Reason
- Source Workstation
- Timestamp

Event ID 4624 – Successful Logon

This event is generated when a user successfully authenticates to the system.

Key fields reviewed:

- Account Name
- Logon Type
- Authentication Package
- Source Information
- Timestamp

Event ID 4634 – Logoff

This event is generated when a user session ends.

Key fields reviewed:

- User Account
- Session Information

- Timestamp

Investigation Findings

Finding 1: Multiple Failed Login Attempts

Several Event ID 4625 entries were observed during the investigation.

Analysis revealed that the user account attempted authentication multiple times using incorrect credentials before a successful login occurred.

Observations:

- Multiple failed authentication attempts were recorded.
- All attempts targeted the same local user account.
- No evidence of remote access was identified.
- The activity originated from the local system.

Potential explanations:

- User typing an incorrect password.
- Password guessing activity.
- Brute-force attempt simulation performed within the lab.

Finding 2: Successful Authentication Following Failed Attempts

A successful login event (Event ID 4624) was recorded after the failed authentication attempts.

Observations:

- Authentication was successful.
- Logon Type indicated an Interactive Logon.
- The login originated from the local machine.
- No suspicious source IP addresses were identified.

Interpretation:

The successful login appears consistent with a legitimate user eventually entering the correct credentials after several failed attempts.

Finding 3: User Session Termination

Event ID 4634 confirmed the termination of the user session.

Observations:

- Logoff activity occurred normally.
- No abnormal session behavior was identified.

Impact Assessment

No evidence of compromise was detected during this investigation.

The observed activity is most likely attributable to normal user behavior within a controlled lab environment. While repeated failed logins can indicate malicious activity in production environments, the context of this investigation suggests the events were generated intentionally for analysis purposes.

Risk Level: Low

Recommendations

1. Monitor repeated failed login attempts to detect potential brute-force attacks.
2. Configure account lockout policies to reduce unauthorized access attempts.
3. Review authentication logs regularly for unusual patterns.
4. Implement centralized log collection for improved visibility and correlation.
5. Continue developing detection capabilities for authentication-related threats.

Conclusion

The investigation successfully identified and analyzed Windows authentication events using native Windows Security Logs. Multiple failed login attempts followed by a successful login were observed and correlated. Based on the evidence collected, the activity was determined to be benign and consistent with expected user behavior in a laboratory environment.

This project demonstrates foundational SOC analyst skills, including event log analysis, authentication monitoring, incident investigation, and security reporting.